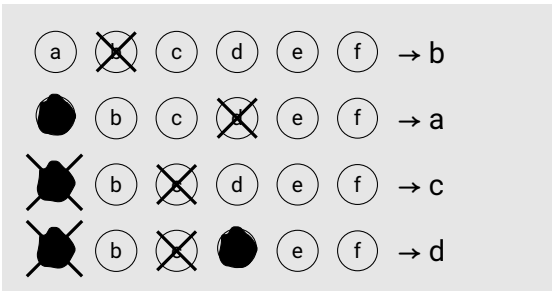
Exercises

1	2	3	4	5	6	7	8

Surname, First name

AI for Medical Imaging (5204AFMI6Y)
22-10-2025 Tentamen
22 Oct, 2025, 14:00-16:00

1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9
0	0	0	0	0	0	0	0



Answer multiple-choice questions as shown in the example.

Fill your name and student number at the top of the page. The student number must be filled twice: spelled-out, and by checking the numbered boxes.

The exam is two hours long (except for students with extra time). One (1) A4 cheatsheet (both sides) is allowed. Phones, calculators and books are disallowed.

For open questions, focus on clarity and concision: the quality of the prose is not graded. The grading scheme is merely indicative and can be changed later.

Modality

1p **1a** Which of the following modality is operator dependent ?

- a Ultrasound
- b MRI
- c CT
- d X-ray
- e PET

1p **1b** Which of the following modality is relatively slow and might need sedation in young children ?

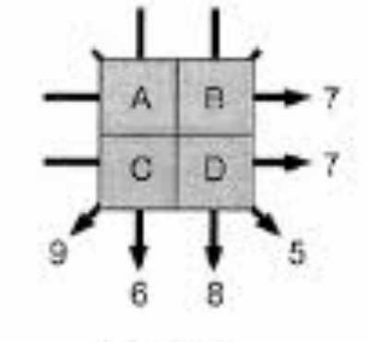
- a Ultrasound
- b MRI
- c CT
- d X-ray
- e PET

1p **1c** What are the two applications of CT imaging ?

2p **1d** A research team participating in the SegTHOR Challenge is developing an automatic segmentation pipeline for lung, heart, trachea, and esophagus from thoracic CT scans. They use a 3D U-Net trained on the provided dataset. During evaluation, they notice that while the model performs well on the lung and heart, the esophagus segmentation has poor overlap with the ground truth. The team suspects that the issue is related to data imbalance and low contrast boundaries in CT images.

Describe one key challenges in segmenting the esophagus from CT scans and one strategies that could improve its performance.

2p **1e** Find the values of A, B, C, and D using the back projection based method for the image shown in the figure below:



Classification

2p **2a** What are the two solutions when you have a large number of classes in a classification problem ?

2p **2b** A hospital is developing a deep learning model to classify chest CT scans as either normal or showing signs of lung cancer. During evaluation, the model achieves high accuracy, but radiologists notice that it often classifies scans with metallic implants or contrast agents as “cancer.”

Why might this misclassification occur, and what modification in the training process could help the model focus on relevant lung features rather than unrelated artifacts?

2p **2c** A research team uses a CNN model to classify chest X-rays as normal or showing pneumonia. To make the predictions explainable, they apply LIME to visualize which image regions influenced the model's decision.
However, they notice that the LIME heatmaps vary widely for the same image when the algorithm is run multiple times, and sometimes highlight non-lung regions like the diaphragm or ribs instead of lung infiltrates.

Why might LIME produce inconsistent or anatomically irrelevant explanations in this case, and how can the reliability of LIME-based interpretations be improved for medical imaging models?

1p **2d** A model predicting disease risk has a baseline output of 0.3.
For one patient, the SHAP values for three features are:
Age: +0.2
BMI: +0.1
Smoking: +0.1
What is the final model prediction for this patient according to SHAP?

(a) 0.3 (b) 0.4 (c) 0.5 (d) 0.6 (e) 0.7

Domain adaptation

3p **3a** A team trains a ResNet-50 model on a large dataset of chest X-rays collected from Hospital A to classify pneumonia vs. normal. When they test the same model on X-rays from Hospital B (which uses a different scanner and contrast settings), the model's accuracy drops sharply from 92% to 68%.
The researchers decide to apply transfer learning and domain adaptation to improve cross-hospital performance.

Why does the model's performance drop when applied to data from Hospital B, and what two techniques from transfer learning or domain adaptation could help improve its generalization?

1p **3b** In a federated learning project, five hospitals collaboratively train a deep learning model for detecting lung nodules on CT scans.

Each hospital trains the model on its own data and sends only the trained model weights (not images) to a central server for aggregation.

After several training rounds, the overall model accuracy is lower than expected, even though each hospital's model performs well locally.
What is the most likely reason for this behavior?

a The hospitals are using the same data, so there is no new information added.

b Each hospital's data comes from different scanners and patient populations, so the combined model struggles to generalize.

c The central server modified the data labels during training.

d The hospitals are not training long enough before sharing updates.

1p **3c** A radiology research group is using the Segment Anything Model (SAM) to segment tumors from MRI scans. They notice that SAM can generate accurate segmentations even when they only draw a few clicks or rough bounding boxes on the image.

What property of SAM allows it to perform segmentation from such minimal input?

a SAM uses prompt-based segmentation, where prompts like points, boxes, or text guide the model to focus on specific regions.

b SAM is trained to predict pixel intensities rather than masks.

c SAM retrains itself every time a user gives a new click or prompt.

d SAM requires a fully annotated training dataset from the same hospital to work correctly.

Reconstruction

4p **4** You are designing a deep learning-based medical imaging pipeline. You can choose between four approaches:

1. Direct DL Method
2. Two-Step / Iterative DL Method
3. Raw-to-Task DL Method
4. Conventional DL Method

You have the following tasks:

Task	Data Quality	Output Requirement	Noise Level	Prior Knowledge
A	High SNR, complete measurements	High-resolution anatomical image	Low	Strong physics model
B	Sparse measurements	Tissue segmentation	Moderate	Limited prior knowledge
C	Limited, noisy measurements	Disease classification	High	Moderate prior knowledge

1. For **Task A**, which DL approach is most suitable? Explain your choice.
2. For **Task B**, which DL approach is most suitable? Explain your choice.
3. For **Task C**, which DL approach is most suitable? Explain your choice.
4. How would computational constraints (e.g., real-time processing) influence your choice for each task?

Segmentation

1p **5a** What are the condition(s) for a loss function to be usable for training?

- (a) Loss value between 0 and 1
- (b) Derivable with respect to the annotation
- (c) Derivable with respect to the network output
- (d) Positive
- (e) Real valued

1p **5b** With a cross-entropy loss defined as $\mathcal{L}_{\text{CE}}(s_\theta, y) \propto \sum_{i\in\Omega}\sum_{k\in\mathcal{K}} y^{(i,k)}\left(-\log(s_\theta^{(i,k)})\right)$, ($\propto$ meaning proportional to), write down the partial derivative of the loss with respect to the predicted probability of voxel i for class k, i.e.: $\frac{\partial \mathcal{L}_{\text{CE}}}{\partial s_\theta^{(i,k)}}$.
Detail your answer.

$-\frac{y^{(i,k)}}{s_\theta^{(i,k)}}$

1p **5c** In which situation can the gradient of the cross-entropy loss go toward infinity? Detail.

$y = 1, \; s_\theta^{(i,k)} \to 0$

1p **5d** Name two methods for post-processing a predicted segmentation. Explain roughly how they work, and when they are applicable or not.

NMS (see Seg II slides)

CRF

1p **5e** A Dice loss function was implemented as follows, but is producing *nan* values for some batches. Suggest a correct fix (writing the new/modified lines is sufficient, no need to rewrite the whole function):

```
def DiceLoss(y: Tensor, s: Tensor) -> Tensor:
    assert y.shape == s.shape
    assert y.ndim > 2
    assert is_onehot(y)
    assert is_simplex(s)

    num = 2 * einsum("bk...,bk...->bk", y, s)
    denom = einsum("bk...->bk", y + s)
    return (1 - num / denom).mean()
```

$\epsilon = 1e-10$

return $(1 - (num + \epsilon)/(denom + \epsilon)).mean()$

1p **5f** You have the following predicted probabilities for a 3 classes problem:

$$s_\theta = \left[\begin{bmatrix} 1 & 0.9 & 0.3 \\ 0.8 & 0.2 & 0.2 \end{bmatrix}, \begin{bmatrix} 0 & 0.1 & 0.6 \\ 0.15 & 0.7 & 0.3 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0.1 \\ 0.05 & 0.1 & 0.5 \end{bmatrix}\right]$$

Write down the predicted segmentation (class numbers) S

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

1p **5g** During one of the guest lecture, SAM and fined-tuned SAM (MedSAM notably) performances were compared on a private CT dataset. Which one performed better?

- (a) SAM
- (b) Med-SAM

Metrics

2p **6a** Re-order the required steps to compute a distance based metric. Some listed below may be surperfluous. Fill the blank with either *boundary* or *area*:

- compute the Euclidean Distance Transform on both ___;
- convert predicted probabilities to one-hot;
- mask distances with the ___;
- reduction of the list of distances into a single number;
- extract the ___;
- upsample for improved precision;

convert to one hot
extract boundary
compute EDT on both boundaries
mask dist w/ boundary
reduction to single number

1p **6b** Fill the blanks (5 to fill):

$\text{HD}(S, Y; k) = \max\left(\max_{i \in \partial\Omega_S^{(k)}} ___{j \in \partial\Omega_Y^{(k)}} d(i,j), ____ d(i,j)\right)$

min max min

2p **6c** You have a task with K classes to segment (+ background). You are required to compute the following metrics:

1. Dice
2. Hausdorff distance
3. ASSD

According to the *textbook* definition, how many times do you need to compute an EDT in total? Motivate your answer.

$2 \times (K+1) \times 2$ → ASSD + HD

EDT in both directions

classes

2p **6d** How could you optimize your metrics computation, and speed things up? Taking into account that the model is not perfect and sometimes does not predict all classes. Explain and illustrate.

- Skip BG :, $k+1 \mapsto k$ } $\Rightarrow 2 \times k$
  reuse EDT across metrics
- If one is empty, skip the EDT & fill with default value

3p **6e** When computing the Hausdorff distance on a *discrete* segmentation (binary setting), explain the two cases where the extracted boundary $\partial\Omega_S$ may be equal to Ω_S (the subset of the image space predicted as foreground). Explain and illustrate.

- line ∂ —— $=$ ——
- Empty $\partial\emptyset = \emptyset$

1p **6f** When benchmarking a 2D fully convolutional network for semantic segmentation, one script reconstructs the 3D volumes from the 2D slices (both for the labels and network predictions). However, due to a bug, the order of the slices during stitching is incorrect (1, 12, 13, 2, 21, 22, ... in stead of 1, 2, 3, ...). The incorrect order is the same between the prediction volume and the label volume.

What is the impact on the computed Dice Score? On the ASSD?

Dice unchanged
ASSD greatly affected (very random/unstable)

Ethics and reproducibility

1p **7a** Rank the following types of images by ease of anonymisation (from the easiest to most difficult). Motivate your answer:

- photo of the iris;
- scan of a biopsy;
- head MRI (including the brain);
- whole body CT;

easy
↓
hard

scan biopsy
head MRI
whole body CT
photo of iris (unique, like fingerprint)

1p **7b** You want to re-use clinical data collected by the hospital, for your research. This requires you to retroactively ask for patient consent. However, some of them are since deceased. What can you do?

Ask ethical board for exemption

1p **7c** Can running the nnU-Net pipeline on a different machine, with a different GPU, find a different "optimal" configuration? Why?

Yes because max memory (and so network size) may change

1p **7d** What is the core difference between anonymisation of the data and *pseudo*-anonymisation?

pseudo anon.: mapping between patient & images is kept somewhere

2p **7e** Define one meaning of *"reproducibility"*, and give a way to improve it.

Ajay:
- Reproduce weights ⇒ random seeds
- Can train on diff machines ⇒ track package versions

1p **7f** What are two main advantages of public challenges for research?

Common benchmark
Reproducible

Registration

1p **8a** What information does the affine matrix of a scan encode?

- (a) rigidity of the patient ???
- (b) size of voxels
- (c) elasticity of the patient ???
- (d) position in the real world
- (e) rotation

1p **8b** You are taking two MRI of a set of patients a few years apart (age at second scan ranging from 13 to 25). The patients moved between the two scans. Depending on which organ you want to compare, what type of (rigid or non-rigid) registration would you expect? Lungs, heart, skull, femur, kidneys. Motivate your choices.

All non-rigid: people grow non linearly

2p **8c** List one advantage and one disadvantage of learned registration over *classical* (optimizer based) registration.

+: fast inference

-: no guarantees of correctness

3p **8d** Assuming you have a U-Net made for image segmentation, what are the minimal architectural changes needed to "convert" it for image registration?

- Take 2 img as input
- Output 3D vectors per voxel & not Δ^{μ} simply
- Use loss for similarity

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